

What is the right geometry for minimization of MMD?

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1 Introduction

Given a functional $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$, its gradient-flow dynamics depend on the choice of geometry on the underlying space of measures. One of the most commonly used geometries is the Wasserstein-2 geometry, induced by the Wasserstein-2 distance. Its associated Onsager operator, which is the inverse of the Riemannian metric tensor, is given by $\mathbb{K}_W(\rho) : T_\rho^* \mathcal{M}^+ \rightarrow T_\rho \mathcal{M}^+, \xi \mapsto -\operatorname{div}(\rho \nabla \xi)$. Hence, the Wasserstein gradient flow of $\mathcal{F}$ is

$$\partial_t \mu_t = \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]). \quad (\text{W-GF})$$

Similarly, the Fisher–Rao geometry is induced by the Hellinger distance, with associated Onsager operator $\mathbb{K}_{\text{FR}}(\rho) : T_\rho^* \mathcal{M}^+ \rightarrow T_\rho \mathcal{M}^+, \xi \mapsto \rho, \xi$. Consequently, the Fisher–Rao gradient flow of $\mathcal{F}$ is

$$\partial_t \mu_t = -\mu_t \left(\mathcal{F}'[\mu_t] - \int \mathcal{F}'[\mu_t] d\mu_t \right). \quad (\text{FR-GF})$$

An interpolation between these two dynamics is given by the well-known Wasserstein–Fisher–Rao gradient flow:

$$\partial_t \mu_t = \alpha \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]) - \beta \mu_t \left(\mathcal{F}'[\mu_t] - \int \mathcal{F}'[\mu_t] d\mu_t \right). \quad (\text{WFR-GF})$$

Recently, a different geometry is considered in [Zhu \[2024\]](#), [Zhu and Mielke \[2024\]](#), [Gladin et al. \[2024\]](#), which uses the MMD as the geometry. The associated Onsager operator is $\mathbb{K}_{\text{MMD}}(\rho) = \mathbb{K}_{\text{FR}}(\rho) \circ \mathcal{K}_\rho^{-1}$. The Interaction-Force Transport (IFT) gradient flow is described as the following:

$$\partial_t \mu_t = -\mathbb{K}_{\text{MMD}}(\mu_t) \mathcal{F}'[\mu_t] = -\mu_t \mathcal{K}_{\mu_t}^{-1} \mathcal{F}'[\mu_t] = -\mathcal{K}^{-1} \mathcal{F}'[\mu_t]. \quad (\text{IFT-GF})$$

Similarly as WFR-GF, an interpolation of WGF and IFT-GF can also be constructed

$$\partial_t \mu_t = \alpha \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]) - \beta \mathcal{K}^{-1} \mathcal{F}'[\mu_t]. \quad (\text{WIFT-GF})$$

2 $\mathcal{F} = \text{MMD}$ and geometry=MMD

We focus on MMD due to its interpretation as the worse-case integration error in the unit ball of a RKHS. If $\mathcal{F}$ is the MMD distance $\mathcal{F}(\cdot) = \text{MMD}^2(\cdot, \pi)$, then $\mathcal{F}'[\mu](\cdot) = \int k(x, \cdot) d(\mu - \pi) = \mathcal{K}(\mu - \pi)$. MSIP is derived from the stationary conditions of an unbalanced WFR gradient flow [\[Belhadji et al., 2026\]](#), whereas here we focus on the WIFT-GF. The main reason is that, [Gladin et al. \[2024\]](#) has proved that $\text{MMD}^2(\mu_t, \pi) \leq e^{-2\beta t} \cdot \text{MMD}^2(\mu_0, \pi)$.

Following Gladin et al. [2024], a way to simulate the WIFT-GF is via an weighted interacting particle system. Let $\mu_N^{(t)} = \sum_{i=1}^N w_i^{(t)} \delta_{x_i^{(t)}}$ be a weighted empirical measure at iteration t . Write $\mathbf{X}_t = (x_1^{(t)}, \dots, x_N^{(t)})$ and $\mathbf{w}^{(t)} = (w_1^{(t)}, \dots, w_N^{(t)})$. Denote $m_\pi = \int k(x, \cdot) d\pi$ as the kernel mean embedding. First evolve the Wasserstein gradient term,

$$x_i^{(t+1)} = x_i^{(t)} - \tau \alpha \left[\sum_{j=1}^N w_j^{(t)} \nabla_2 k(x_j^{(t)}, x_i^{(t)}) - \nabla m_\pi(x_i^{(t)}) \right]. \quad (1)$$

Then, evolve the IFT gradient term, by solving the following optimal weights: let $\eta = \beta \tau$,

$$\mathbf{w}^{(t+1)} = \arg \min_{\mathbf{w} \in \Delta^N} \left\{ \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i^{(t+1)}}, \pi \right) + \frac{1}{2\eta} \text{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i^{(t+1)}}, \mu_N^{(t)} \right) \right\}. \quad (2)$$

Interestingly, letting $K^{(t+1)} = K(\mathbf{X}_{t+1}) \in \mathbb{R}^{N \times N}$, the weight problem is

$$\min_{\mathbf{w} \in \Delta^N} \left\{ \frac{1}{2} \mathbf{w}^\top K^{(t+1)} \mathbf{w} - m_\pi(\mathbf{X}_{t+1})^\top \mathbf{w} + \frac{1}{2\eta} (\mathbf{w} - \mathbf{w}^{(t)})^\top K^{(t+1)} (\mathbf{w} - \mathbf{w}^{(t)}) \right\},$$

By definition of kernel quadrature, there is $m_\pi(\mathbf{X}_{t+1}) = K^{(t+1)} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$. And the solution is, when ignoring the simplex constraint,

$$\mathbf{w}^{(t+1)} = \frac{1}{1 + \eta} \mathbf{w}^{(t)} + \frac{\eta}{1 + \eta} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}).$$

To state the particle updates compactly, define the kernel-mean residual

$$h_{\mathbf{X}, \mathbf{w}}(x) := \sum_{j=1}^N w_j k(x_j, x) - m_\pi(x).$$

Table 1 gives the location and weight updates explicitly for the three geometries and for MSIP [Belhadji et al., 2026]. In the WFR scheme, the transport step precedes the weight step, which is evaluated at the updated particle locations. The WFR weight update is obtained by applying forward Euler discretization to the Fisher–Rao reaction term in Eq. (WFR-GF).

The Wasserstein row uses uniform initial weights, which remain equal to $1/N$ under transport; arbitrary initial weights would likewise remain fixed. The centering term in the WFR update preserves $\sum_{i=1}^N w_i^{(t)} = 1$, while nonnegativity additionally requires $\tau\beta$ to be sufficiently small. The MMD row gives the closed-form weight update obtained when the simplex constraint in Eq. (2) is ignored. The WIFT discretization in Eqs. (1)–(2) combines the Wasserstein location update with the MMD weight update in Table 1. In contrast, MSIP is a damped fixed-point method for the stationary conditions of the unbalanced WFR flow, rather than a time discretization of the normalized WFR flow in Eq. (WFR-GF). Its update requires $K(\mathbf{X}_t)$ to be invertible and the kernel quadrature weights to be positive.

In contrast, the update scheme for the standard Euclidean gradient is:

$$x_i^{(t+1)} = x_i^{(t)} - \tau \alpha w_i \left[\sum_{j=1}^N w_j^{(t)} \nabla_2 k(x_j^{(t)}, x_i^{(t)}) - \nabla m_\pi(x_i^{(t)}) \right], \quad \mathbf{w}^{(t+1)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}). \quad (3)$$

Comparing the two update schemes, we note that the two only differs by a weight factor w_i .

Table 1: Location and weight updates for MMD minimization under Wasserstein, WFR, and MMD geometries, together with MSIP.

Geometry or method	Location and weight updates
Wasserstein	$x_i^{(t+1)} = x_i^{(t)} - \tau \alpha \nabla h_{\mathbf{X}_t, \mathbf{w}^{(t)}}(x_i^{(t)}),$ $w_i^{(t+1)} = w_i^{(t)} = \frac{1}{N}.$ $x_i^{(t+1)} = x_i^{(t)} - \tau \alpha \nabla h_{\mathbf{X}_t, \mathbf{w}^{(t)}}(x_i^{(t)}),$
WFR	$w_i^{(t+1)} = w_i^{(t)} - \tau \beta w_i^{(t)} \left[h_{\mathbf{X}_{t+1}, \mathbf{w}^{(t)}}(x_i^{(t+1)}) - \sum_{j=1}^N w_j^{(t)} h_{\mathbf{X}_{t+1}, \mathbf{w}^{(t)}}(x_j^{(t+1)}) \right].$ $x_i^{(t+1)} = x_i^{(t)},$
MMD	$\mathbf{w}^{(t+1)} = \frac{1}{1 + \eta} \mathbf{w}^{(t)} + \frac{\eta}{1 + \eta} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}), \quad \mathbf{X}_{t+1} = \mathbf{X}_t.$
MSIP	For a squared-exponential kernel, $x_i^{(t+1)} = (1 - \tau) x_i^{(t)} + \frac{\tau}{w_{\text{KQ}, i}(\mathbf{X}_t)} \sum_{j=1}^N [K(\mathbf{X}_t)^{-1}]_{ij} \int x k(x, x_j^{(t)}) d\pi(x),$ $\mathbf{w}^{(t+1)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}).$

Definition 2.1 (Weighted Euclidean Geometry). *Let $m \in \mathbb{R}$ and let $G \in \mathbb{R}^{m \times m}$ be a symmetric positive definite matrix. We equip the space $\mathbb{R}^m$ with the G -weighted Euclidean inner product $\langle u, v \rangle_G := u^\top G v$ and with the induced norm $\|u\|_G^2 := u^\top G u$. For a differentiable function $f : \mathbb{R}^m \rightarrow \mathbb{R}$, its gradient with respect to the G -weighted Euclidean geometry, denoted by $\text{grad}_G f$, is defined as: for any $u \in \mathbb{R}^m$,*

$$\left. \frac{d}{dt} f(x + tu) \right|_{t=0} = \langle \text{grad}_G f(x), u \rangle_G.$$

Equivalently, $\text{grad}_G f(x) = G^{-1} \nabla f(x)$ where ∇f denotes the standard Euclidean gradient.

Proposition 2.2 (Smoothness under Euclidean and weighted Euclidean geometries). *Let $\mathbf{w} \in \Delta^N$ with $w_i > 0$ for every i and $\sum_{i=1}^N w_i = 1$. Suppose that k is symmetric and twice continuously differentiable, with $\sup_{x, y \in \mathbb{R}^d} \|\nabla_{11}^2 k(x, y)\|_{\text{op}} = C_1 < \infty$ and $\sup_{x, y \in \mathbb{R}^d} \|\nabla_{12}^2 k(x, y)\|_{\text{op}} = C_2 < \infty$. Define*

$$E(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i}, \pi \right).$$

Then $E(\cdot, \mathbf{w})$ is $\max_{1 \leq i \leq N} w_i (2C_1 + C_2)$ -smooth under the standard Euclidean geometry; and is $(2C_1 + C_2)$ -smooth under the $\mathbf{w}$ -weighted Euclidean geometry. Moreover, $E(\mathbf{X}, \cdot)$ is $\lambda_{\max}(K(\mathbf{X}, \mathbf{X}))$ -smooth under the standard Euclidean geometry; and is 1-smooth under the $K(\mathbf{X}, \mathbf{X})$ -weighted Euclidean geometry.

From the above proposition, we see that the right smoothness notion to consider is the weighted Euclidean geometry instead of the standard Euclidean geometry.

The direct KQ update instantaneously minimizes the MMD over the weights at each particle configuration, whereas the IFT update is a proximal tracking scheme that interpolates between the previous weights and the instantaneous KQ solution. The latter sacrifices instantaneous optimality in exchange for a controlled and stable evolution of the weights, consistent with the underlying continuous-time IFT dynamics.

3 Convergence

Theorem 3.1 (Monotone improvement). *At iteration $t \in \mathbb{N}$, let $\mu_N^{(t)} = \sum_{i=1}^N w_i^{(t)} \delta_{x_i^{(t)}}$ with $\mathbf{X}_t = (x_1^{(t)}, \dots, x_N^{(t)})$, and $\mathbf{w}^{(t)} = (w_1^{(t)}, \dots, w_N^{(t)})$.*

Assume that k is twice continuously differentiable, that $K(\mathbf{X}_t)$ is positive definite for every t , and that the weights remain uniformly positive, namely $w_i^{(t)} \geq \underline{w} > 0$ for every i, t . Assume moreover that, for every admissible $\mathbf{w}$, the map $\mathbf{X} \mapsto \mathcal{E}(\mathbf{X}, \mathbf{w})$ has an L_X -Lipschitz gradient on a set containing all iterates. If $\tau\alpha \leq 2\underline{w}/L_X$, then for every $t \geq 0$,

$$\text{MMD}(\mu_N^{(t+1)}, \pi) \leq \text{MMD}(\mu_N^{(t)}, \pi). \quad (4)$$

Lemma 3.2 (Uniform control of the kernel quadrature residual). *Suppose that the assumptions of Theorem 3.1 hold and that the initial weights are chosen as $\mathbf{w}^{(0)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$. Then, for every $t \geq 0$, $\text{MMD}(\sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_t) \delta_{x_i^{(t)}}, \pi) \leq \text{MMD}(\sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_0) \delta_{x_i^{(0)}}, \pi)$.*

Theorem 3.3 (Convergence to the kernel quadrature error). *Suppose that the assumptions of Theorem 3.1 hold, and that the initial weights are chosen as $\mathbf{w}^{(0)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$. Then, the WIFT iterates satisfy*

$$\text{MMD}^2(\mu_N^{(t)}, \pi) \leq \frac{1}{(1+\eta)^{2t}} \text{MMD}^2(\mu_N^{(0)}, \pi) + \left(1 - \frac{1}{(1+\eta)^{2t}}\right) \text{MMD}^2\left(\sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_0) \delta_{x_i^{(0)}}, \pi\right).$$

Remark 3.4. *Theorem 3.3 separates the optimization error from the statistical error of the initial kernel quadrature rule. In particular, if $\text{MMD}(\sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_0) \delta_{x_i^{(0)}}, \pi) = O_{\mathbb{P}}(N^{-s/d})$, then*

$$\text{MMD}(\mu_N^{(t)}, \pi) \leq \frac{1}{(1+\eta)^t} \text{MMD}(\mu_N^{(0)}, \pi) + O_{\mathbb{P}}(N^{-s/d}).$$

Thus, the optimization error decays geometrically until it reaches the kernel quadrature error floor. If, in addition, $\text{MMD}(\mu_N^{(0)}, \pi) = O_{\mathbb{P}}(1)$, then choosing $t \geq \frac{s}{d \log(1+\eta)} \log N$ is sufficient to obtain $\text{MMD}(\mu_N^{(t)}, \pi) = O_{\mathbb{P}}(N^{-s/d})$.

4 Proofs

Proof of Proposition 2.2. Recall that

$$E(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \sum_{i,j=1}^N w_i w_j k(x_i, x_j) - \sum_{i=1}^N w_i m_{\pi}(x_i) + \text{const.}$$

For any $\mathbf{u} = (u_1, \dots, u_N) \in (\mathbb{R}^d)^N$, direct differentiation gives

$$\begin{aligned}\nabla_{\mathbf{X}}^2 E(\mathbf{X}, \mathbf{w})[\mathbf{u}, \mathbf{u}] &= \sum_{i,j=1}^N w_i w_j u_i^\top \nabla_{11}^2 k(x_i, x_j) u_i \\ &\quad + \sum_{i,j=1}^N w_i w_j u_i^\top \nabla_{12}^2 k(x_i, x_j) u_j - \sum_{i=1}^N w_i u_i^\top \nabla^2 m_\pi(x_i) u_i.\end{aligned}$$

Since $m_\pi(x) = \int k(x, y) d\pi(y)$ and π is a probability measure, we have

$$\|\nabla^2 m_\pi(x)\|_{\text{op}} = \left\| \int \nabla_{11}^2 k(x, y) d\pi(y) \right\|_{\text{op}} \leq \sup_{x, y \in \mathbb{R}^d} \|\nabla_{11}^2 k(x, y)\|_{\text{op}} = C_1.$$

Using $\sum_{j=1}^N w_j = 1$, it follows that

$$\left| \sum_{i,j=1}^N w_i w_j u_i^\top \nabla_{11}^2 k(x_i, x_j) u_i - \sum_{i=1}^N w_i u_i^\top \nabla^2 m_\pi(x_i) u_i \right| \leq 2C_1 \sum_{i=1}^N w_i \|u_i\|^2.$$

For the mixed term,

$$\left| \sum_{i,j=1}^N w_i w_j u_i^\top \nabla_{12}^2 k(x_i, x_j) u_j \right| \leq \sup_{x, y \in \mathbb{R}^d} \|\nabla_{12}^2 k(x, y)\|_{\text{op}} \left(\sum_{i=1}^N w_i \|u_i\| \right)^2 = C_2 \left(\sum_{i=1}^N w_i \|u_i\| \right)^2.$$

By Cauchy–Schwarz and $\sum_{i=1}^N w_i = 1$, we have $\left(\sum_{i=1}^N w_i \|u_i\| \right)^2 \leq \sum_{i=1}^N w_i \|u_i\|^2$. Therefore,

$$|\nabla_{\mathbf{X}}^2 E(\mathbf{X}, \mathbf{w})[\mathbf{u}, \mathbf{u}]| \leq (2C_1 + C_2) \sum_{i=1}^N w_i \|u_i\|^2 = (2C_1 + C_2) \|\mathbf{u}\|_{\mathbf{w}}^2.$$

The last step holds by the definition of $\mathbf{w}$ -weighted Euclidean geometry in Definition 2.1. This proves smoothness under the weighted Euclidean geometry.

For the standard Euclidean geometry, since $\sum_{i=1}^N w_i \|u_i\|^2 \leq (\max_{1 \leq i \leq N} w_i) \sum_{i=1}^N \|u_i\|^2 = (\max_{1 \leq i \leq N} w_i) \|\mathbf{u}\|_2^2$. The proof is concluded. $\square$

Proof of Theorem 3.1. For fixed $\mathbf{X}$ and $\mathbf{w}$, let $g_i(\mathbf{X}, \mathbf{w}) = \sum_{j=1}^N w_j \nabla_2 k(x_j, x_i) - \nabla m_\pi(x_i)$. Define the MMD energy by

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i}, \pi \right), \quad (5)$$

Differentiating the MMD energy with respect to x_i and using symmetry of the kernel gives $\nabla_{x_i} \mathcal{E}(\mathbf{X}, \mathbf{w}) = w_i g_i(\mathbf{X}, \mathbf{w})$. Hence, the position update can be written as $x_i^{(t+1)} = x_i^{(t)} - \frac{\tau \alpha}{w_i^{(t)}} \nabla_{x_i} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$. Thus, the Wasserstein substep is a diagonally preconditioned gradient step for $\mathbf{X} \mapsto \mathcal{E}(\mathbf{X}, \mathbf{w}^{(t)})$. By the L_X -smoothness of $\mathcal{E}(\cdot, \mathbf{w}^{(t)})$, the descent lemma gives

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) + \sum_{i=1}^N \left\langle \nabla_{x_i} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}), x_i^{(t+1)} - x_i^{(t)} \right\rangle + \frac{L_X}{2} \sum_{i=1}^N \left\| x_i^{(t+1)} - x_i^{(t)} \right\|^2$$

$$= \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) - \tau\alpha \sum_{i=1}^N \left(w_i^{(t)} - \frac{L_X \tau\alpha}{2} \right) \left\| g_i(\mathbf{X}_t, \mathbf{w}^{(t)}) \right\|^2. \quad (6)$$

Since $w_i^{(t)} \geq \underline{w}$ and $\tau\alpha \leq 2\underline{w}/L_X$, we have $w_i^{(t)} - L_X \tau\alpha/2 \geq 0$. Therefore, $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$. This proves that the particle update in Eq. (1) is decreasing the MMD value.

We next consider the weight update. For fixed $\mathbf{X}$, the dependence of the MMD energy on $\mathbf{w}$ is quadratic: $\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \mathbf{w}^\top K(\mathbf{X}) \mathbf{w} - m_\pi(\mathbf{X})^\top \mathbf{w} + \text{const}$. Since $K(\mathbf{X})$ is positive definite, the unique minimizer is $\mathbf{w}_{\text{KQ}}(\mathbf{X}) = K(\mathbf{X})^{-1} m_\pi(\mathbf{X})$. Completing the square gives

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \mathcal{E}(\mathbf{X}, \mathbf{w}_{\text{KQ}}(\mathbf{X})) + \frac{1}{2} (\mathbf{w} - \mathbf{w}_{\text{KQ}}(\mathbf{X}))^\top K(\mathbf{X}) (\mathbf{w} - \mathbf{w}_{\text{KQ}}(\mathbf{X})). \quad (7)$$

By the weight update,

$$\mathbf{w}^{(t+1)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) = \frac{1}{1+\eta} \left(\mathbf{w}^{(t)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) \right). \quad (8)$$

On the other hand, for fixed particle locations $\mathbf{X}_{t+1}$, the KQ decomposition gives

$$\begin{aligned} & \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\ &= \frac{1}{2} \left(\mathbf{w}^{(t+1)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) \right)^\top K^{(t+1)} \left(\mathbf{w}^{(t+1)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) \right). \end{aligned} \quad (9)$$

Substituting (8) into (9) yields

$$\begin{aligned} & \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\ &= \frac{1}{2(1+\eta)^2} \left(\mathbf{w}^{(t)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) \right)^\top K^{(t+1)} \left(\mathbf{w}^{(t)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) \right) \\ &= \frac{1}{(1+\eta)^2} \left[\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \right]. \end{aligned} \quad (10)$$

Since $(1+\eta)^{-2} \leq 1$ and $\mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$ minimizes $\mathbf{w} \mapsto \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w})$, it follows that $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)})$. Combining the two substeps yields

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}). \quad (11)$$

Hence, $\mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$ is nonincreasing, and the proof is concluded. $\square$

Proof of Lemma 3.2. Recall the definition that $\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i}, \pi \right)$. For any fixed particle locations $\mathbf{X}_t$, the kernel quadrature weights $\mathbf{w}_{\text{KQ}}(\mathbf{X}_t)$ minimize the MMD energy over the weights. Hence, $\mathcal{E}(\mathbf{X}_t, \mathbf{w}_{\text{KQ}}(\mathbf{X}_t)) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$. On the other hand, by the monotonicity of the WIFT iteration established in Theorem 3.1, we have $\mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}^{(0)})$. Since $\mathbf{w}^{(0)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$, we have $\mathcal{E}(\mathbf{X}_0, \mathbf{w}^{(0)}) = \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0))$. Combining the previous inequalities gives

$$\mathcal{E}(\mathbf{X}_t, \mathbf{w}_{\text{KQ}}(\mathbf{X}_t)) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}^{(0)}) = \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)). \quad (12)$$

The proof is thus concluded. $\square$

Proof of Theorem 3.3. From the exact contraction of the weight update in Eq. (10), together with descent of the Wasserstein position update, we have

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \frac{1}{(1+\eta)^2} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) + \left(1 - \frac{1}{(1+\eta)^2}\right) \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})). \quad (13)$$

By Lemma 3.2, $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0))$, and therefore

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \frac{1}{(1+\eta)^2} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) + \left(1 - \frac{1}{(1+\eta)^2}\right) \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)). \quad (14)$$

The proof is concluded. $\square$

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